Enterprise GIS and the City

an inductive, autoethnographic analysis of working as a geospatial analyst supporting local government

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City of Cedar Rapids
WLAAG 2026/10/17

AAG
American Association of Geographers

ILLINOIS

Cedar Rapids
City of Five Seasons®

Outline & Abstract

Background
- Geography
- Cedar Rapids
- Information Technology

What is Enterprise GIS
- Privileges, administration
- versioning, services

Project Anecdotes
- Project Tracking
- Street Storage
- Infrastructure Viewer
- Snowplow Tracks

Conclusions
- Collaboration
- Workflow Analysis
- Continuous Learning

What is an Enterprise GIS, and why should you care? This presentation draws upon two years of work experience as a geospatial analyst supporting local government to help answer these questions.

More formally, I aim to provide an autoethnographic reflection of several real-world projects from ideation to implementation. In particular, this reflection should make clear the distinctions between an Enterprise GIS and common classroom GIS environments.

Ultimately, perhaps even more important than any particular technological skills, my reflections identify the professional skills that appear to be most valued by local government entities: collaboration, workflow design, and continuous learning.

AAG
American Association of Geographers

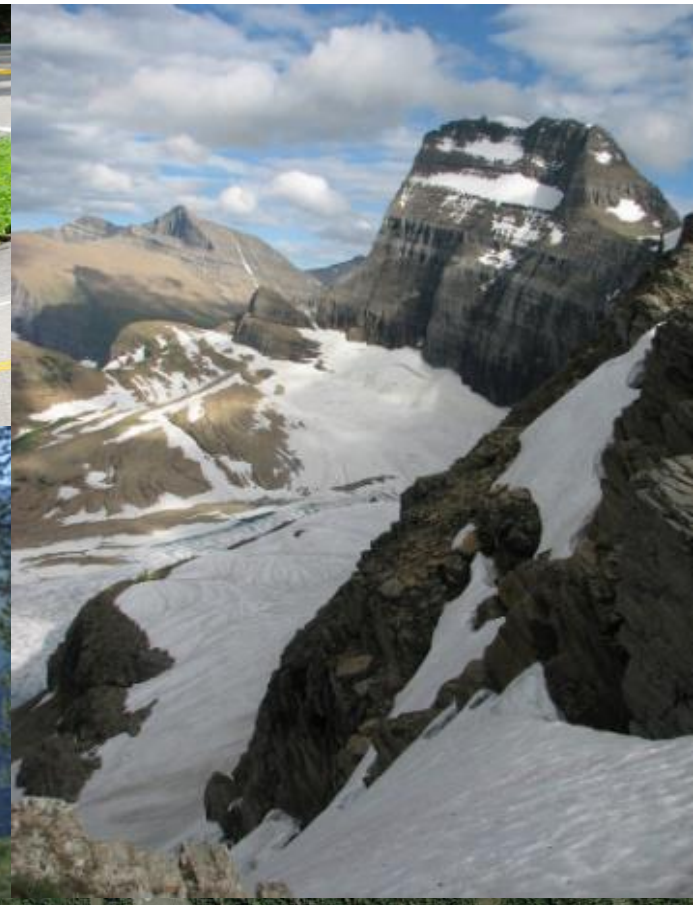
ILLINOIS

CEDAR RAPIDS
City of Five Seasons®

Background: *a geographer*















Geospatial Analyst, *current*
City of Cedar Rapids

PhD @ Texas State U., 2022:
Bicycle Level of Service

MA @ U. of Northern Iowa, 2018:
Risk Perception in Automotive Environments

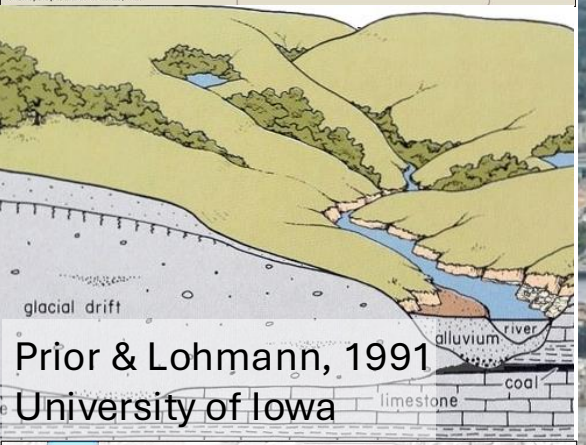
Photogrammetrist, 2017
Aerial Services, Inc.

GIS Data Coordinator, 2015
Iowa Dept. of Transportation

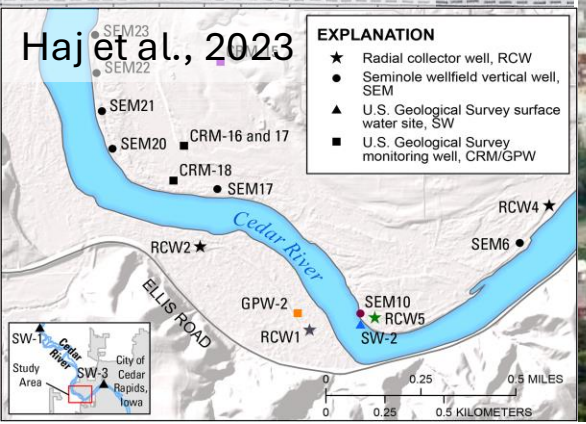
Background: City of Cedar Rapids



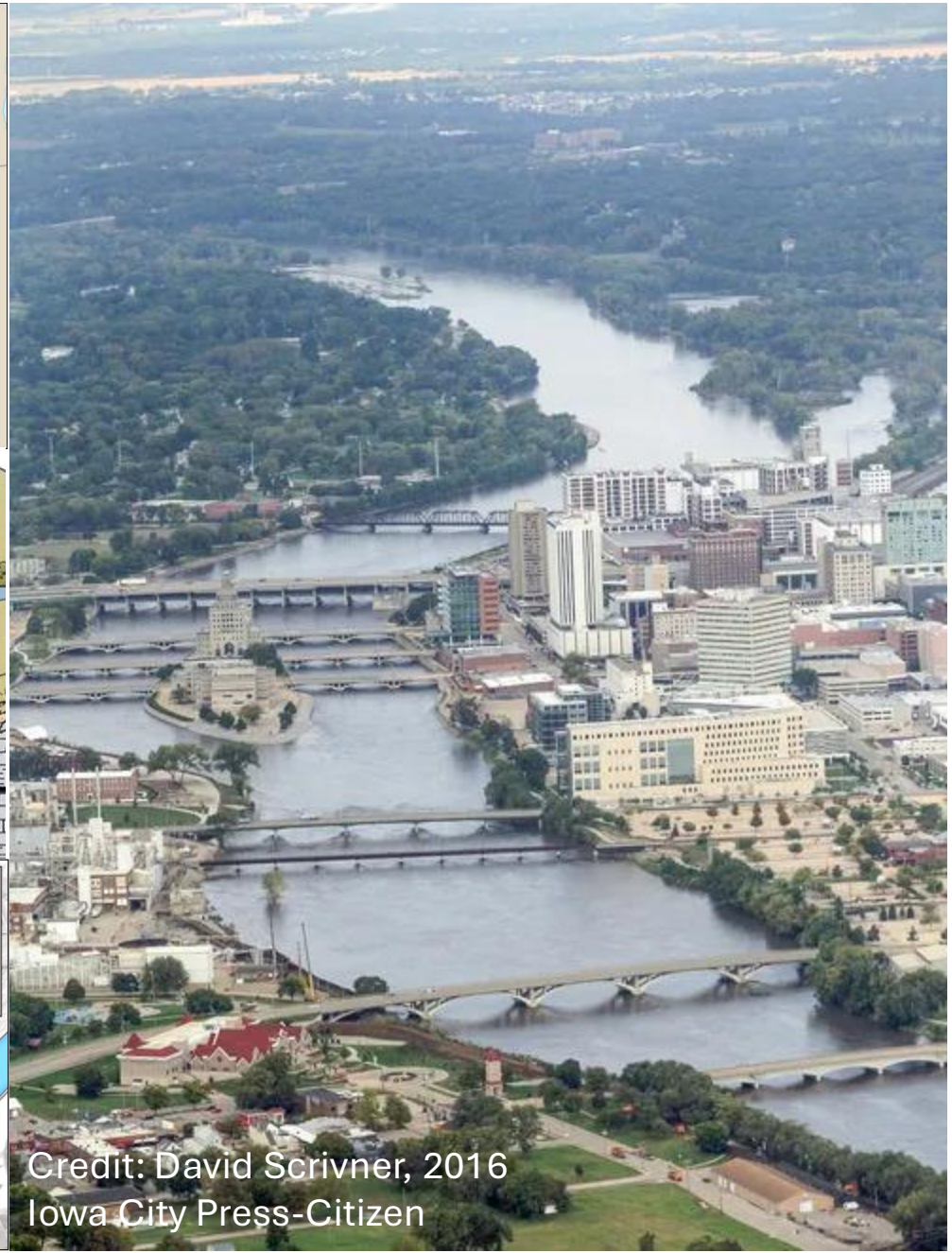
Brittanica, 2026



Prior & Lohmann, 1991
University of Iowa



Haj et al., 2023


Credit: David Scrivner, 2016
Iowa City Press-Citizen

On the Cedar River,
Straddling the Southern Iowa Drift Plain and the Iowan (Erosional) Surface.

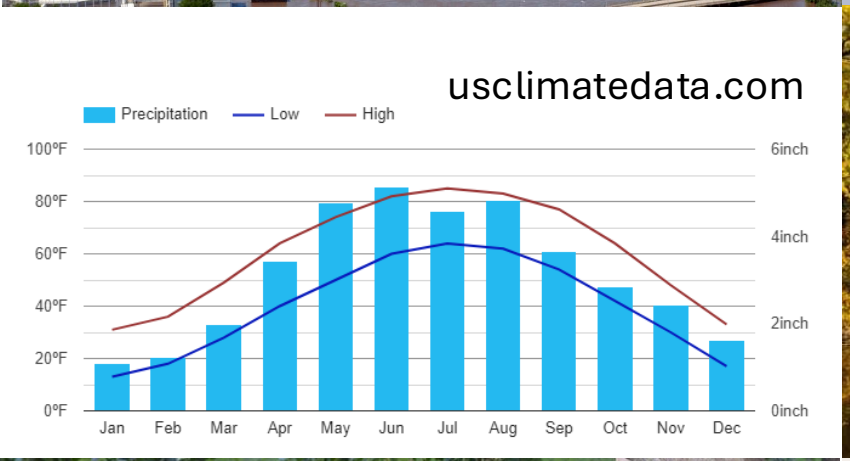
Background: City of Cedar Rapids





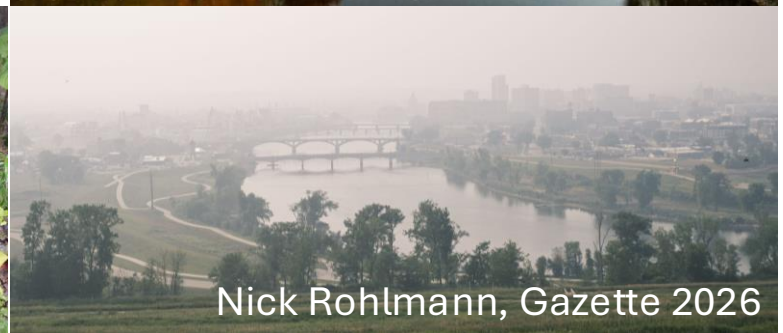
On the Cedar River,
Straddling the Southern Iowa Drift Plain and the Iowan (Erosional) Surface.

(Dfa) Humid continental, several months of snow cover yet still hot summers.









Background: City of Cedar Rapids



Travel Iowa


Hello Cedar Rapids





Ben Kaplan


Benjamin Hill, MLB.com

On the Cedar River,
Straddling the Southern Iowa Drift Plain and the Iowan (Erosional) Surface.

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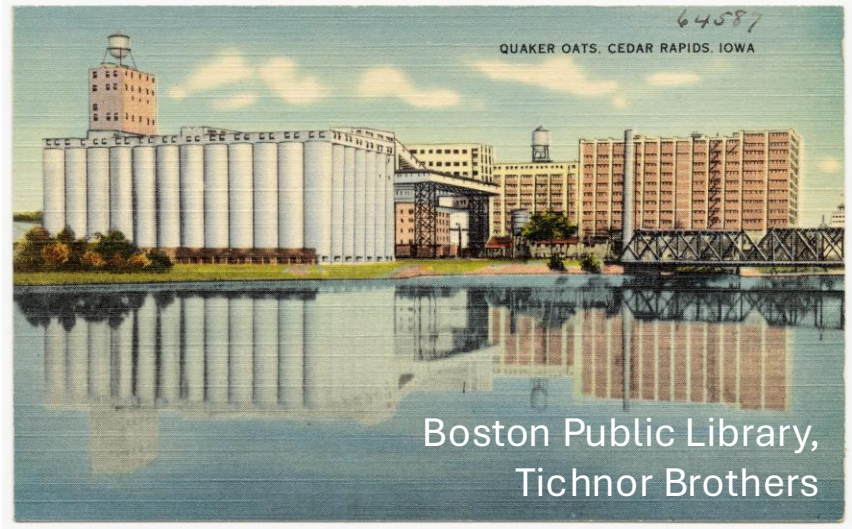
2nd Largest City in Iowa,
135,000 Residents,
~250,000 in the Metro

Background: City of Cedar Rapids


Josh Scheinblum, KCRG



Jackie Kennon, KCRG



Boston Public Library,
Tichnor Brothers



Corridor Business Journal

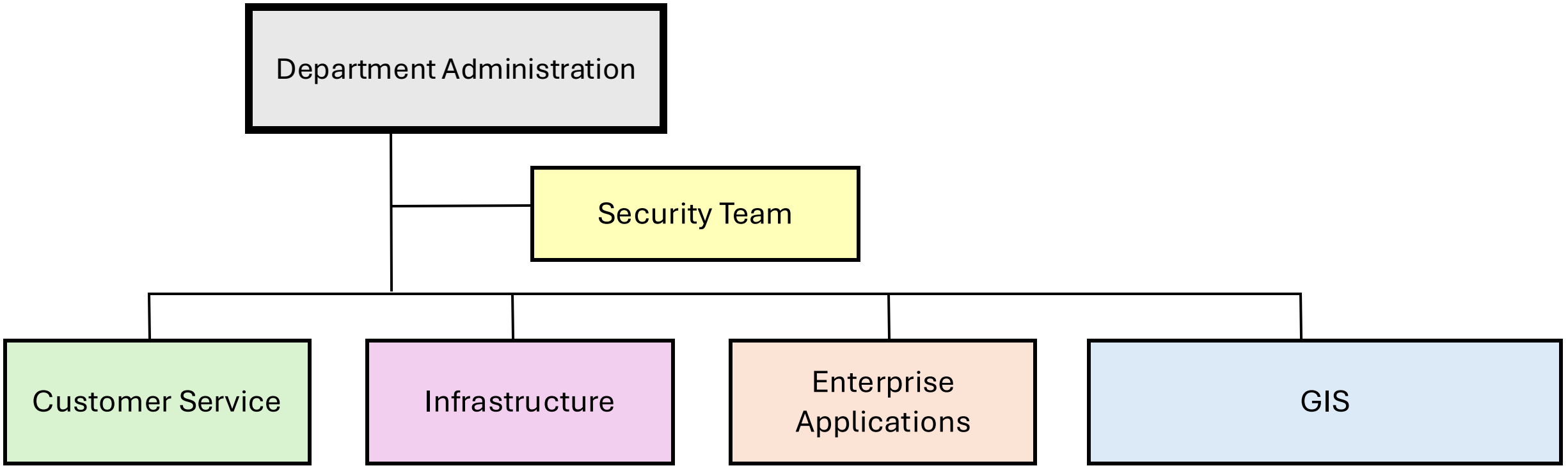
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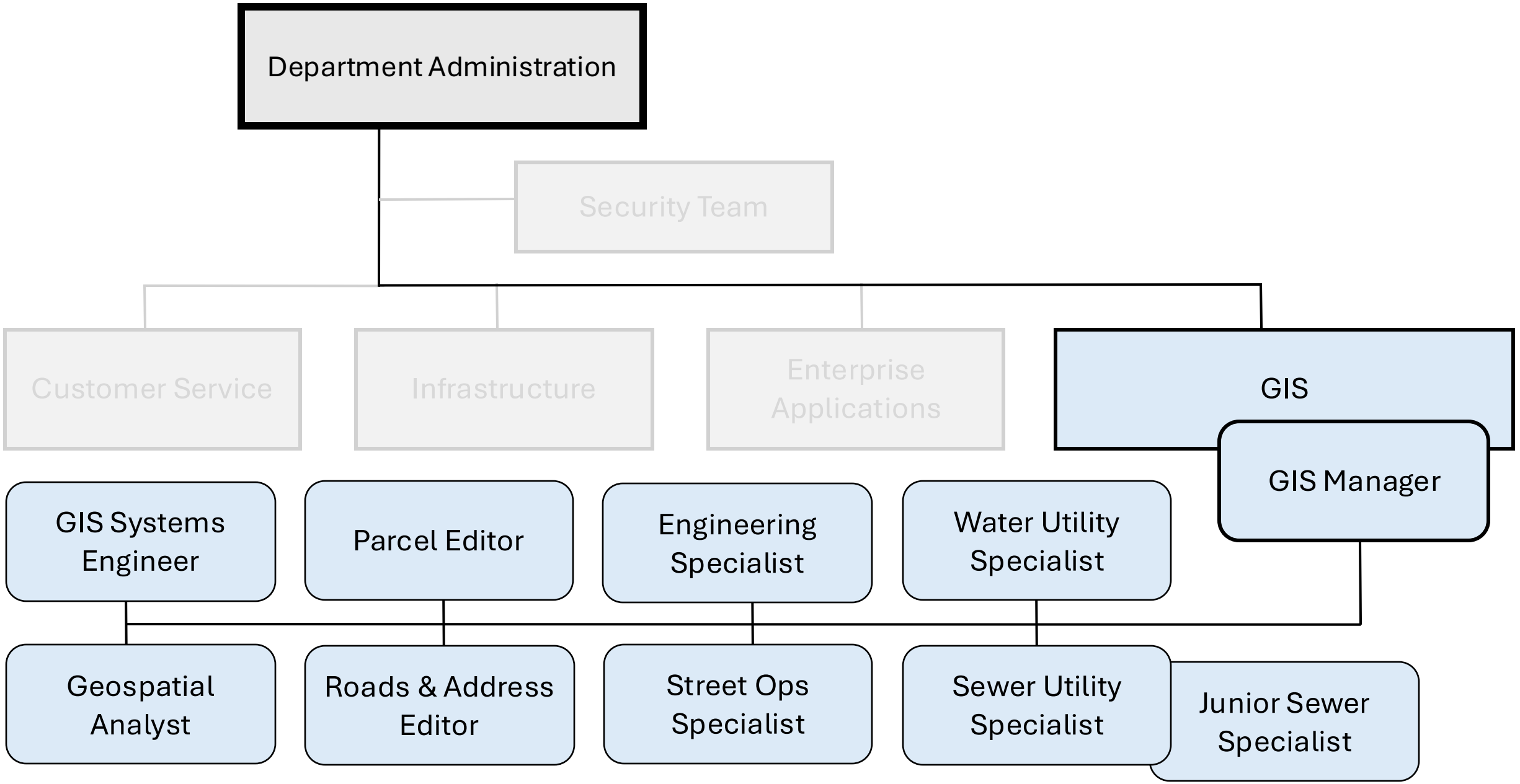
2nd Largest City in Iowa,
135,000 Residents,
~250,000 in the Metro

Major manufacturing and aerospace Industries; Quaker Oats, Collins, … agricultural supply chain.

Background: Information Technology

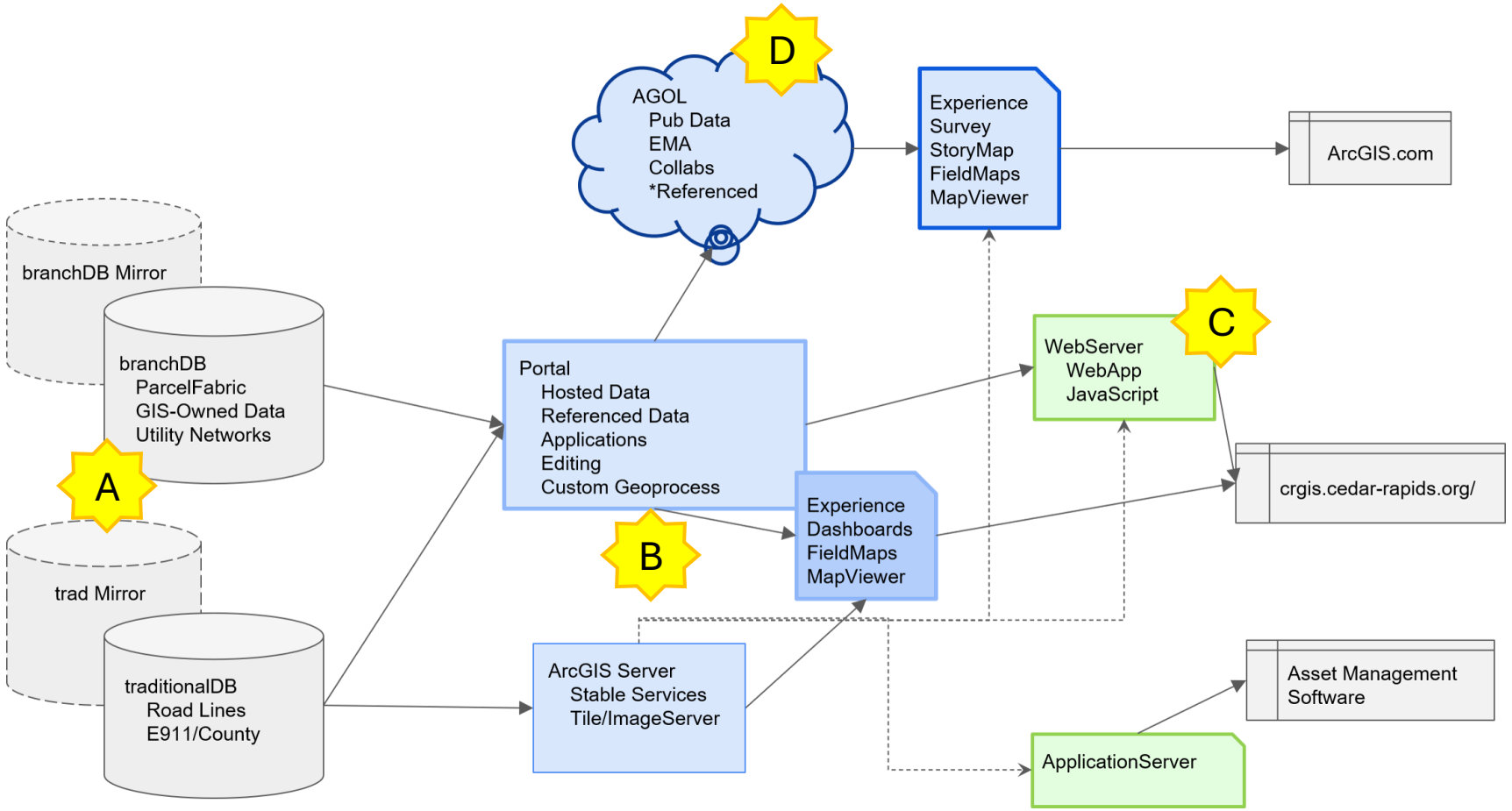


Background: Information Technology



What is *Enterprise GIS?*

- [information systems] A geographic information system that is integrated through an entire organization so that a large number of users can manage, share, and use spatial data and related information to address a variety of needs, including data creation, modification, visualization, analysis, and dissemination.



Cedar Rapids GIS Architecture

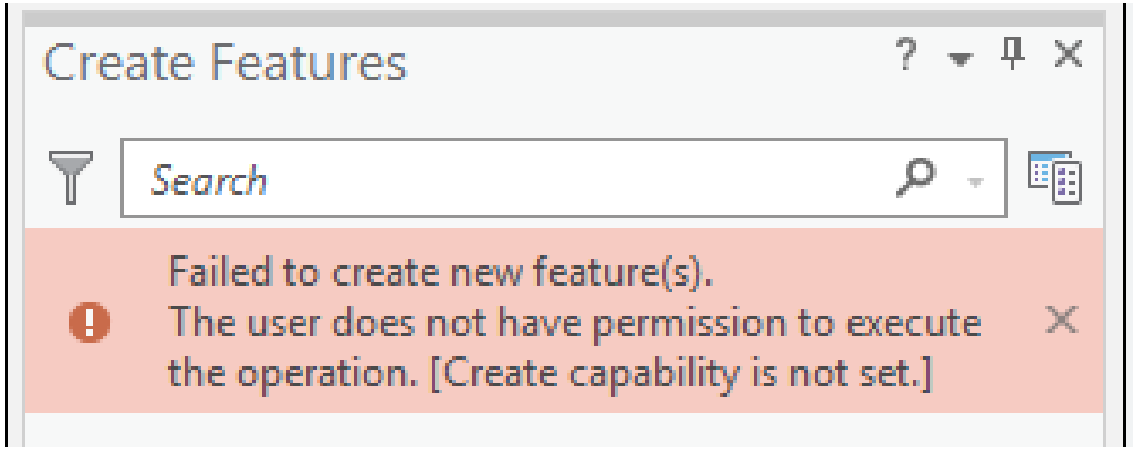
A Database:
4 SQL Servers

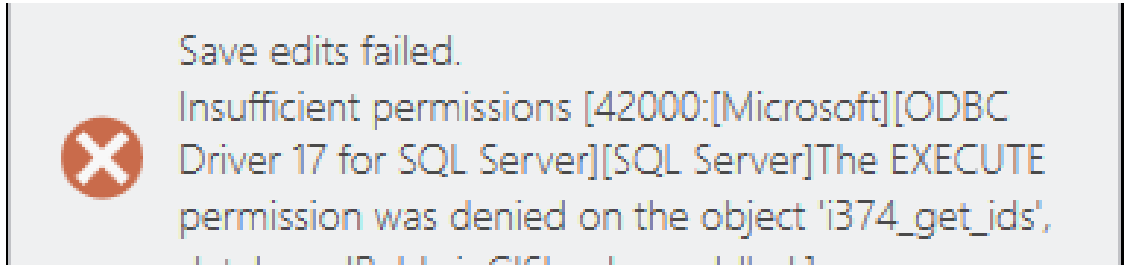
B GIS Servers:
3 Portal Servers
2 ArcGIS Servers

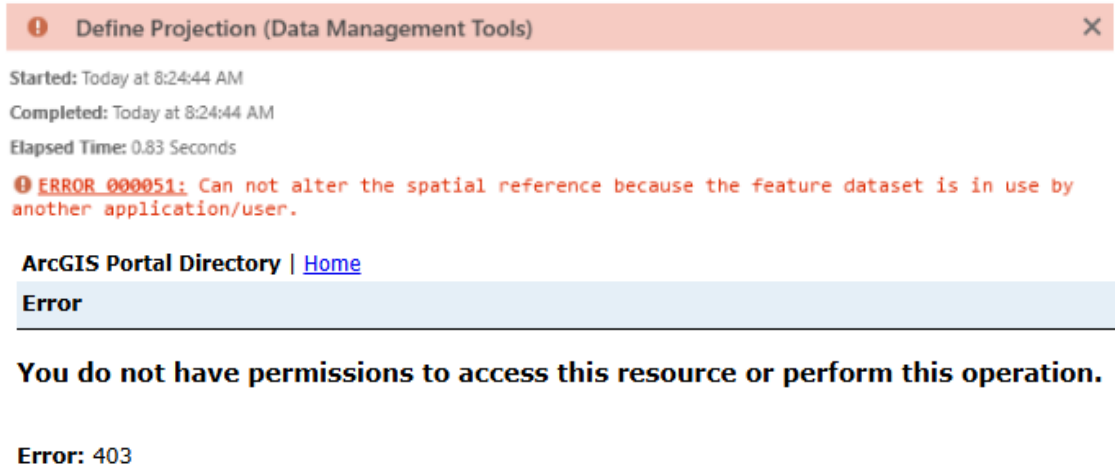
C WebServer:
1 Windows Server/IIS

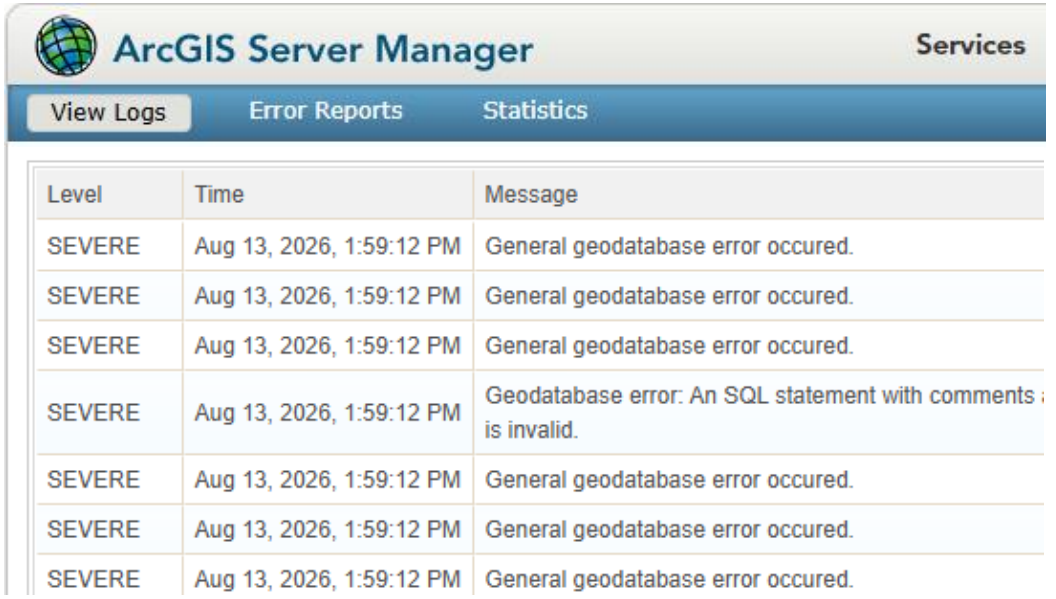
D ArcGIS Online Cloud Env.

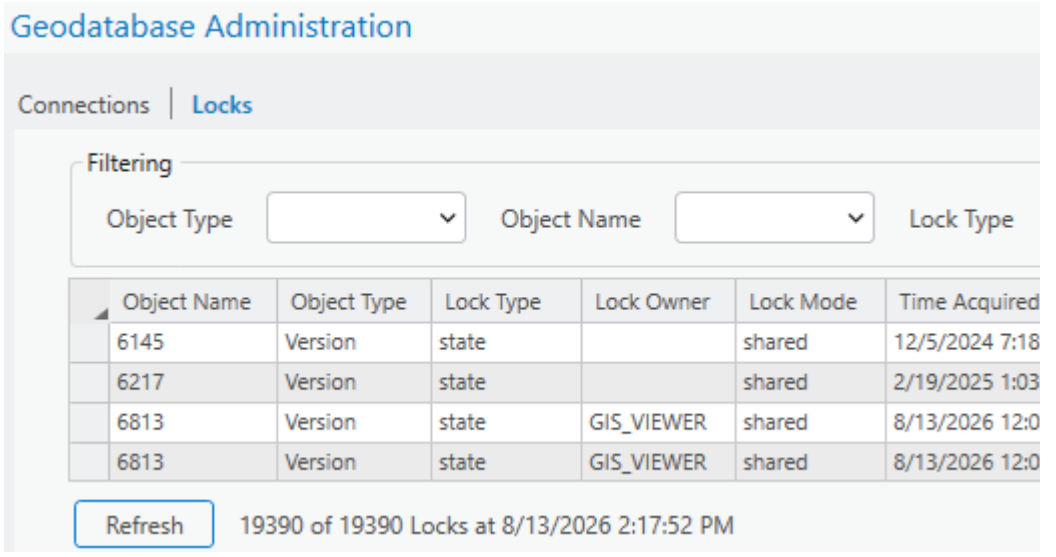
What is *Enterprise GIS?*



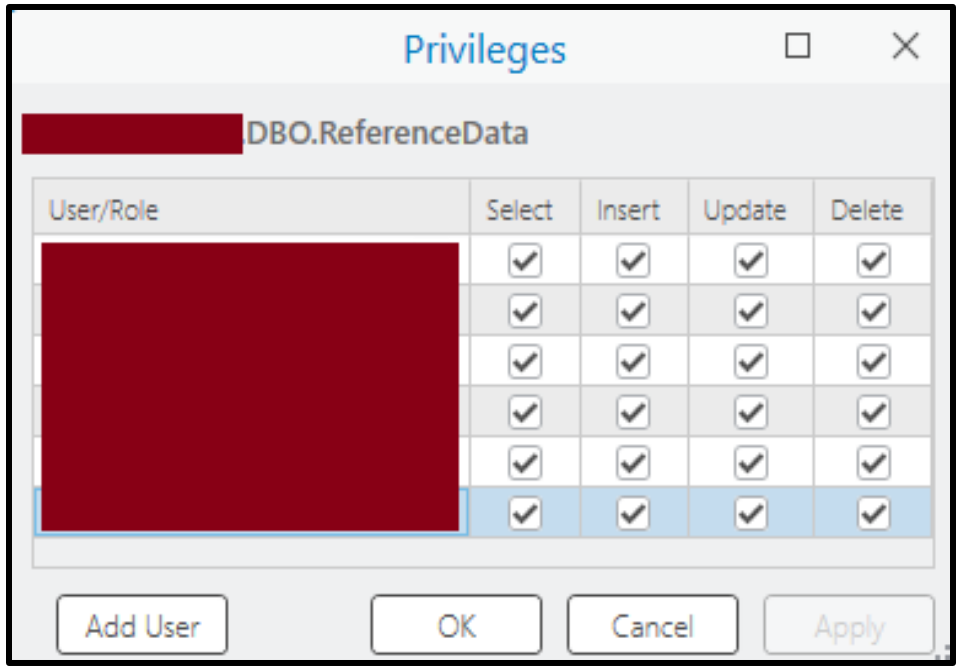


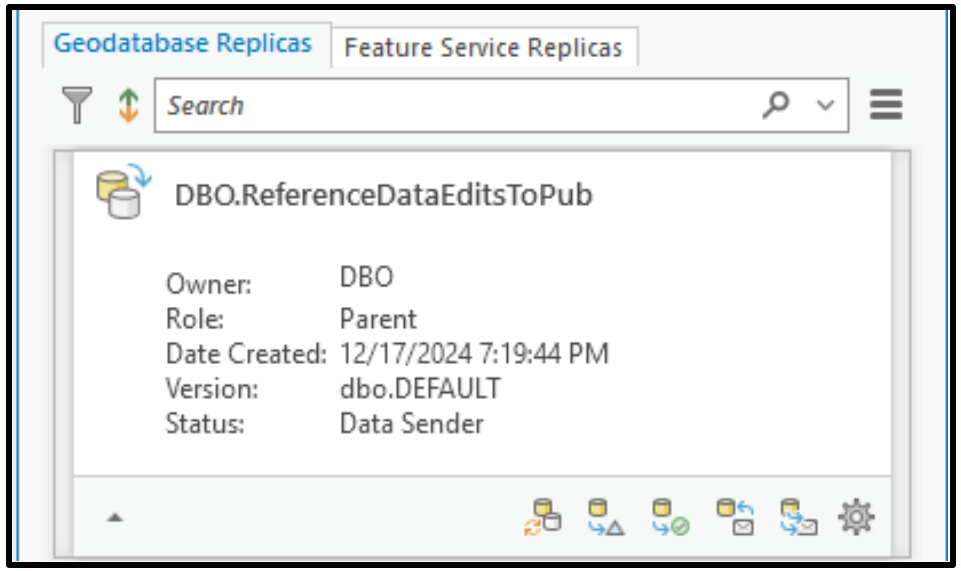


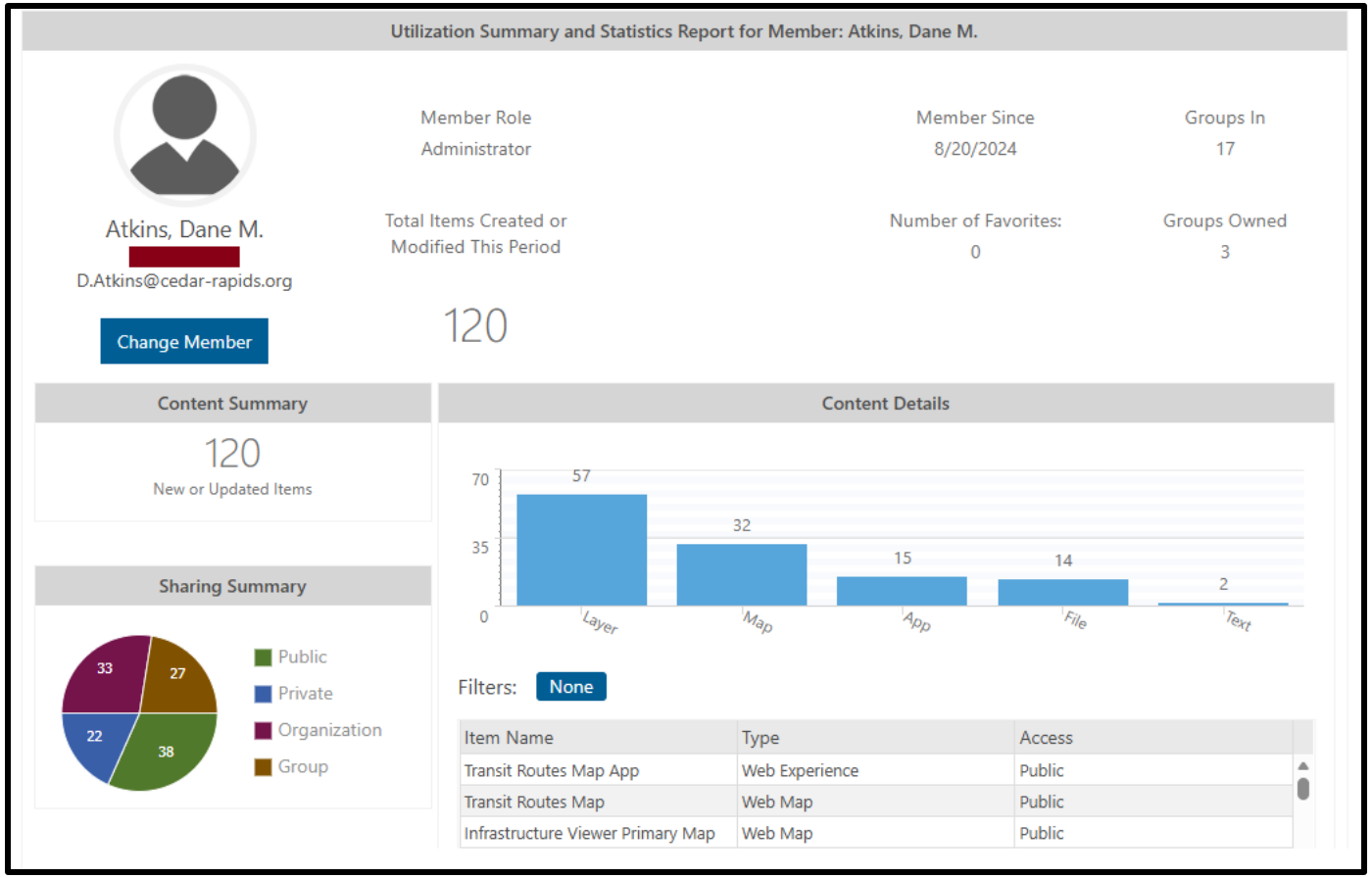




What is *Enterprise GIS?*



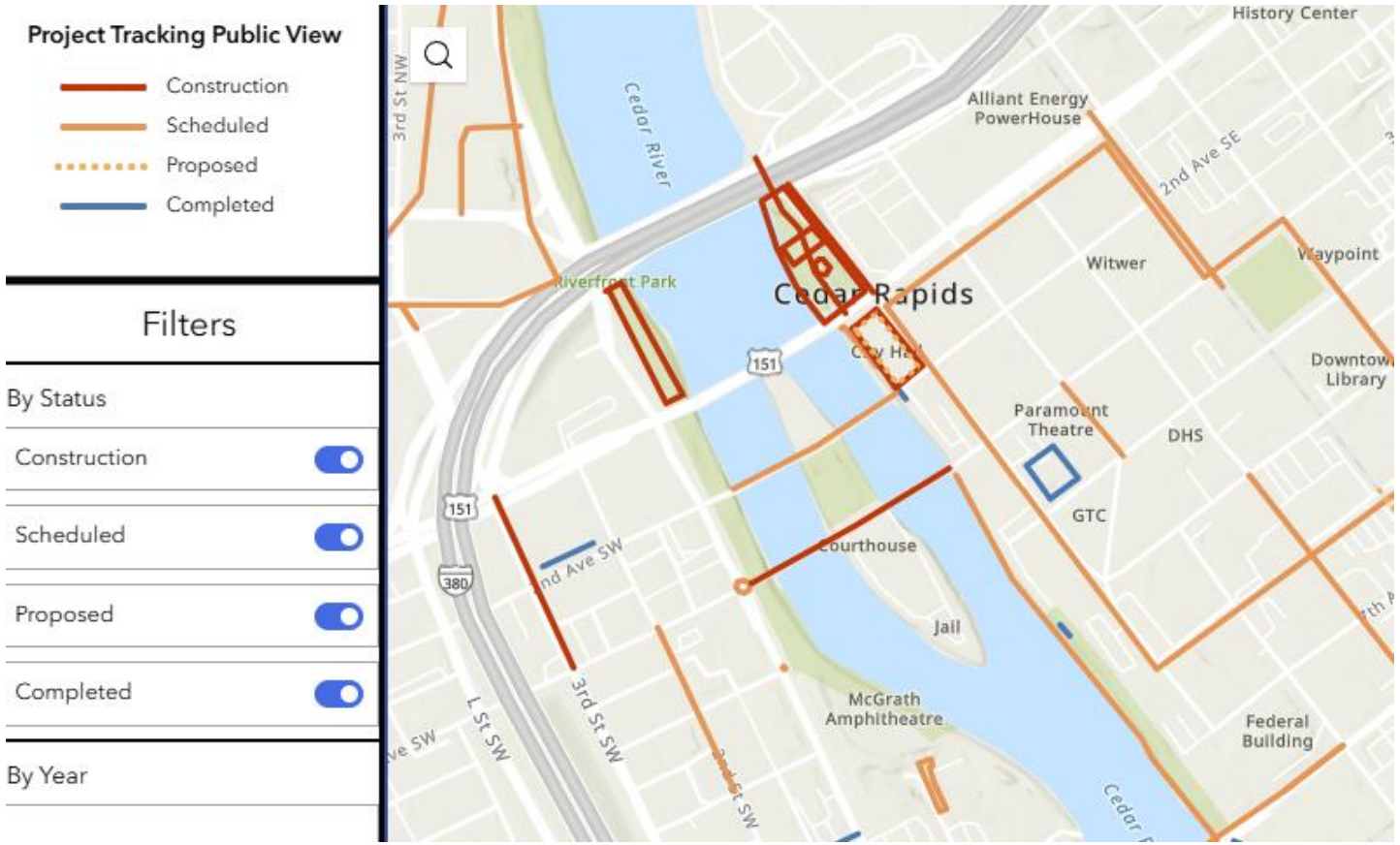


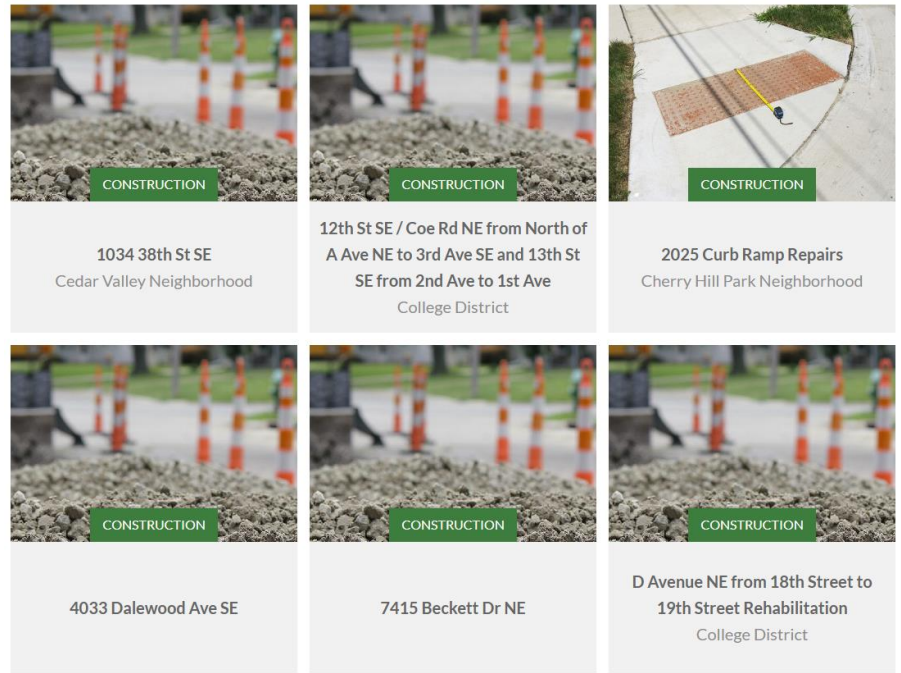




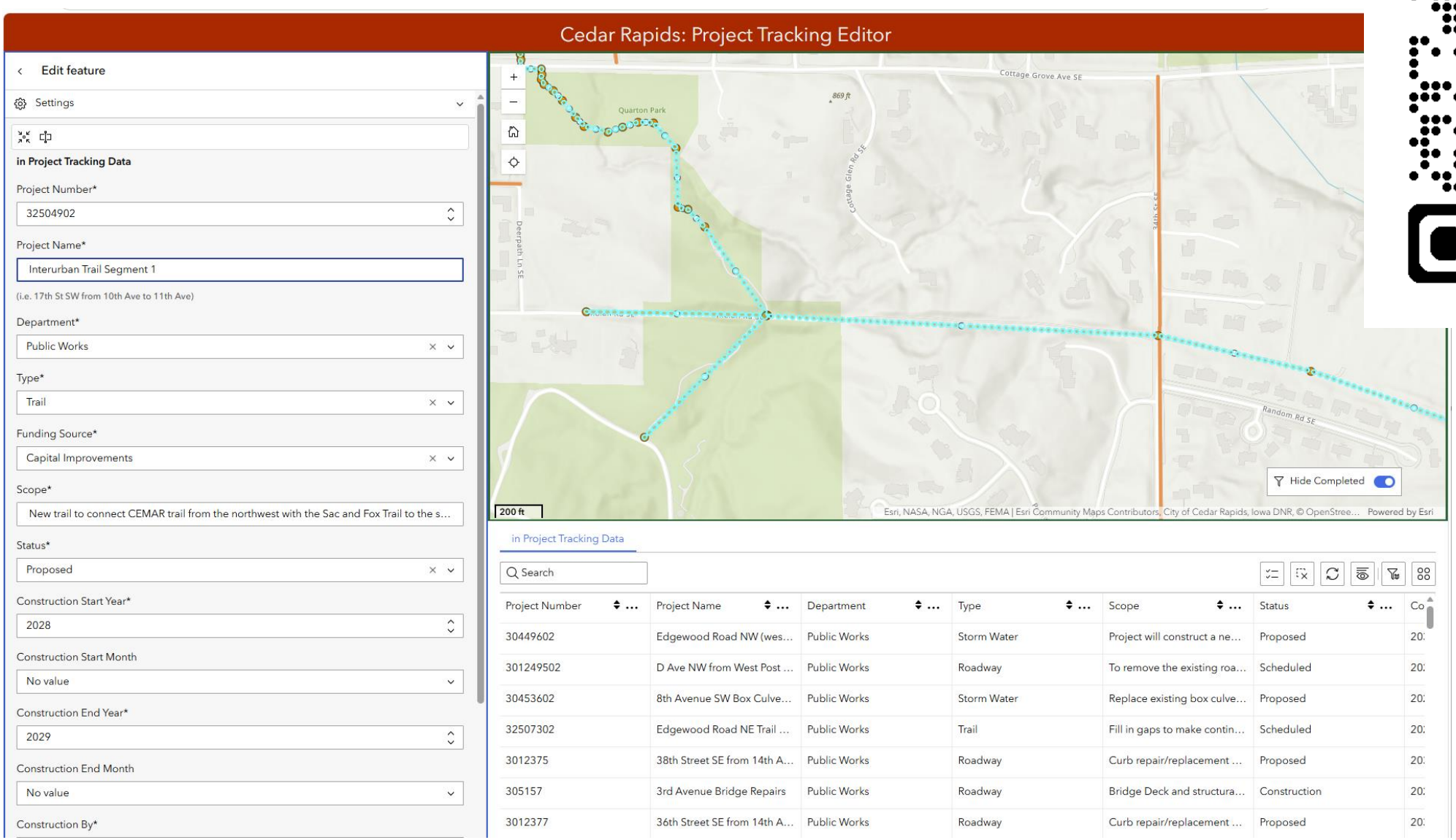
… and why should you care?

Capital Improvements' Project Tracking





Capital Improvements' Project Tracking

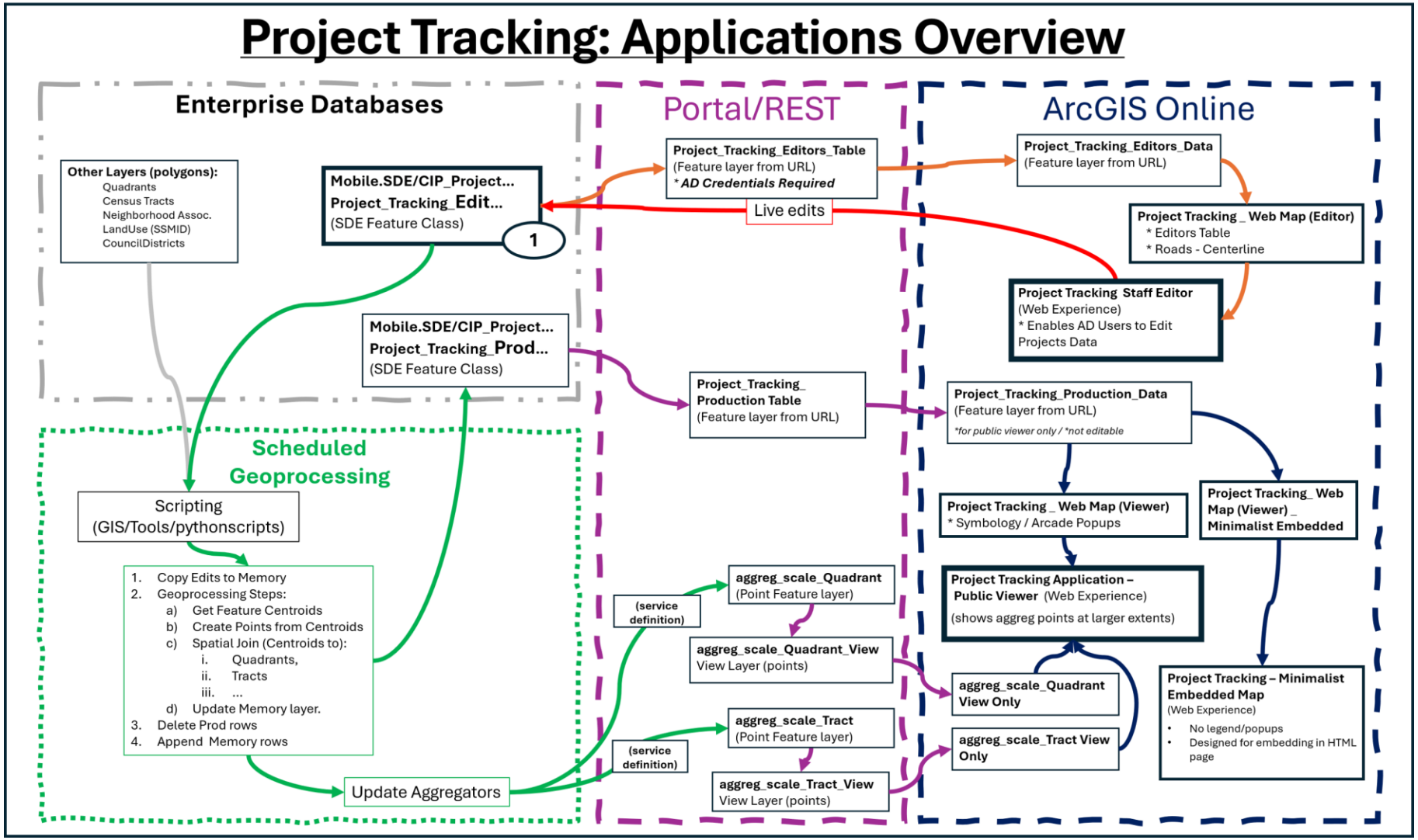


Capital Improvements' Project Tracking

...collaboration



Capital Improvements' Project Tracking




```
##########################
##### FUNCTION CALLS #####
##########################
halt = False

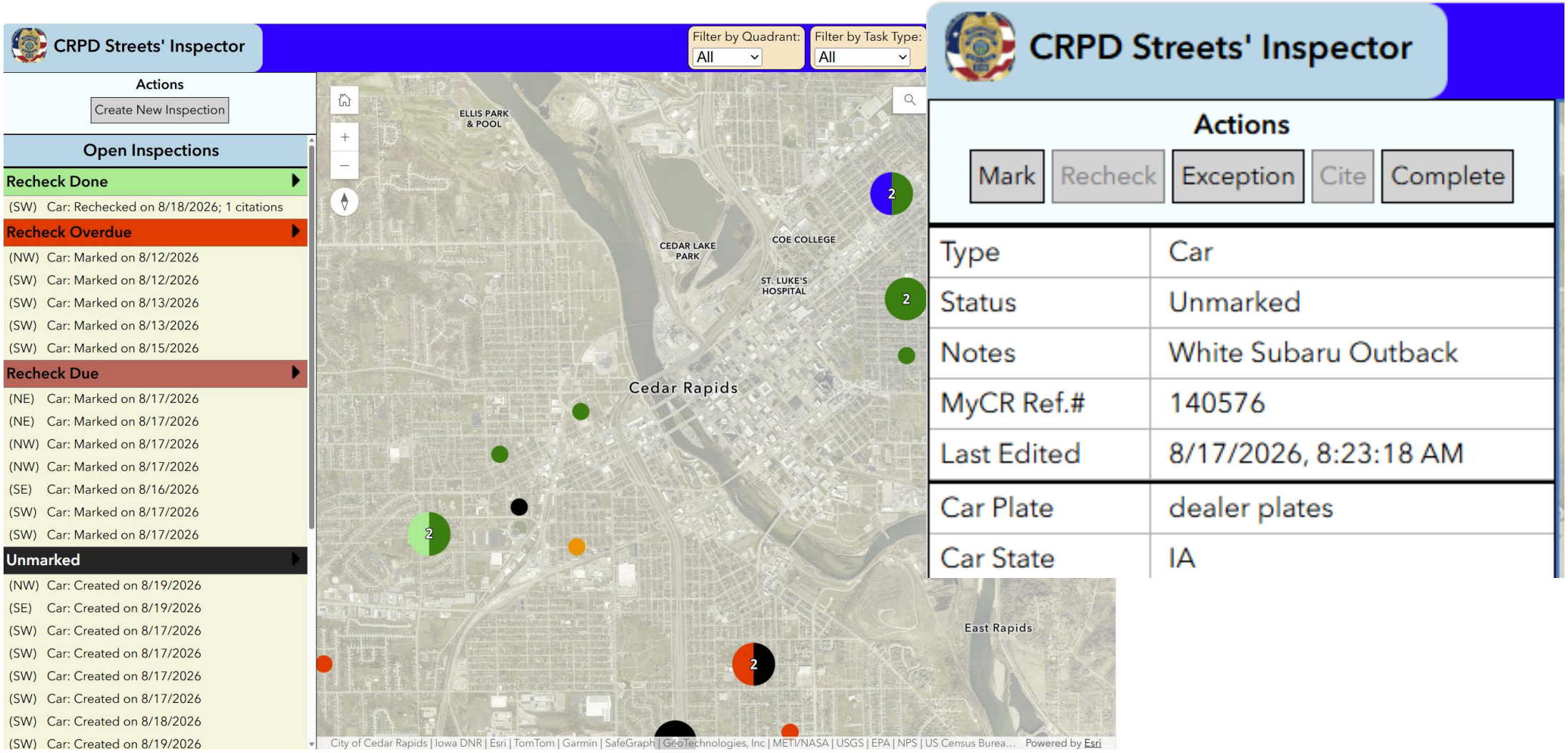
try:

    initializations()
except Exception as e:
    halt = True
    msg("Critical Error during initializations.")
    msg(str(e))
    arc_log.error("Fatal exception as initializations()")
    arc_log.post("Failed", "initializations()")
    arc_log.email(to_list = ['d.atkins@cedar-rapids.org',

# Optional: Update Report
if not halt:
    try:
        update_report()
    except:
        msg("    ...WARNING: Failed to generate an Update
        arc_log.error("Failed to create a Project Update
        pass

if not halt:
    # REQUIRED: Basic Updates (copy from table to memory)
    try:
        msg("...Starting update procedures...")
        queue_edits()
    except Exception as e:
        halt = True
        msg("!!! INCIDENT WARNING: Basic updates failed.
```

Street Storage Inspector



Street Storage Inspector

...Workflow Analysis

Create New Inspection

Inputs:

Location	*(map-point)*
Type	*(selector)*
Comment	*(text)*
MyCR#	*(number)*
(if 'Car' Type):	
Plate	*(text)*
State	*(selector)*

Autofill:

Status	*('Unmarked')*
Address	*(from Location)*
District	*('')*
Quadrant	*('')*
createdDate	*(current DateTime)*

Mark

Inputs:

MarkedComment	*(text)*

Autofill:

Status	*'Marked'*
MarkedBy	*(preselected)*
MarkedDate	*(current)*
editedDate	*(database)*

Recheck Due / Overdue

Inputs:

RecheckComment	*(text)*

Autofill:

Status	*'Recheck'*
RecheckBy	*(preselected)*
RecheckDate	*(current)*
editedDate	*(database)*

Complete

Inputs:

CompleteComment	*(comment)*
CompleteMethod	*(selector)*
(if 'Towed' Method):	
Citation4	*(text)*
CaseNo	*(text)*

Autofill:

CompleteBy	*(preselected)*
CompleteDate	*(selector)*
editedDate	*(database)*

Exception

	(Yes/No)
ment	*(text)*
ate	*(date)*
	'Exception'
	(database)

Cite #1

Inputs:

Citation1	*(text)*

Autofill:

Citation1Date	*(current)*
Citation1By	*(preselected)*
editedDate	*(database)*

Cite #2

Inputs:

Citation2	*(text)*

Autofill:

Citation2Date	*(current)*
Citation2By	*(preselected)*
editedDate	*(database)*

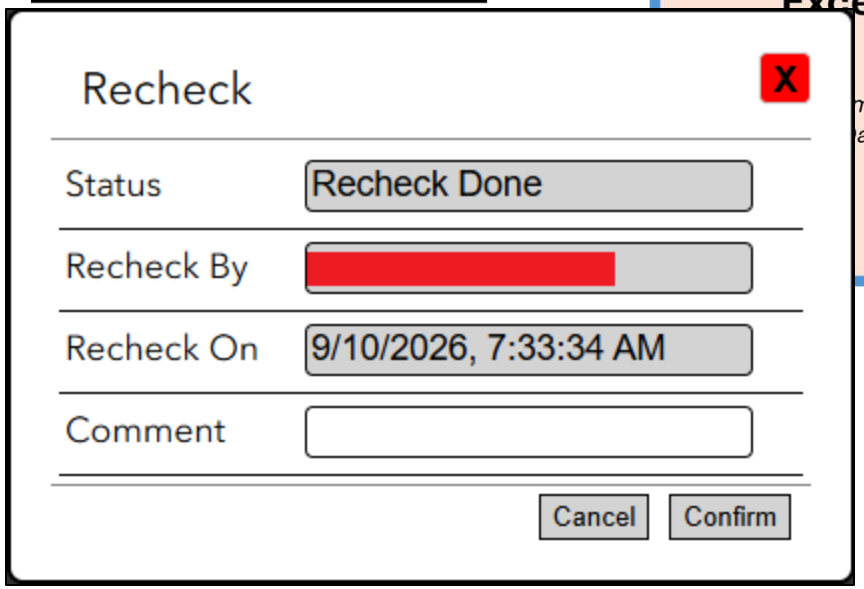
Cite #3

Inputs:

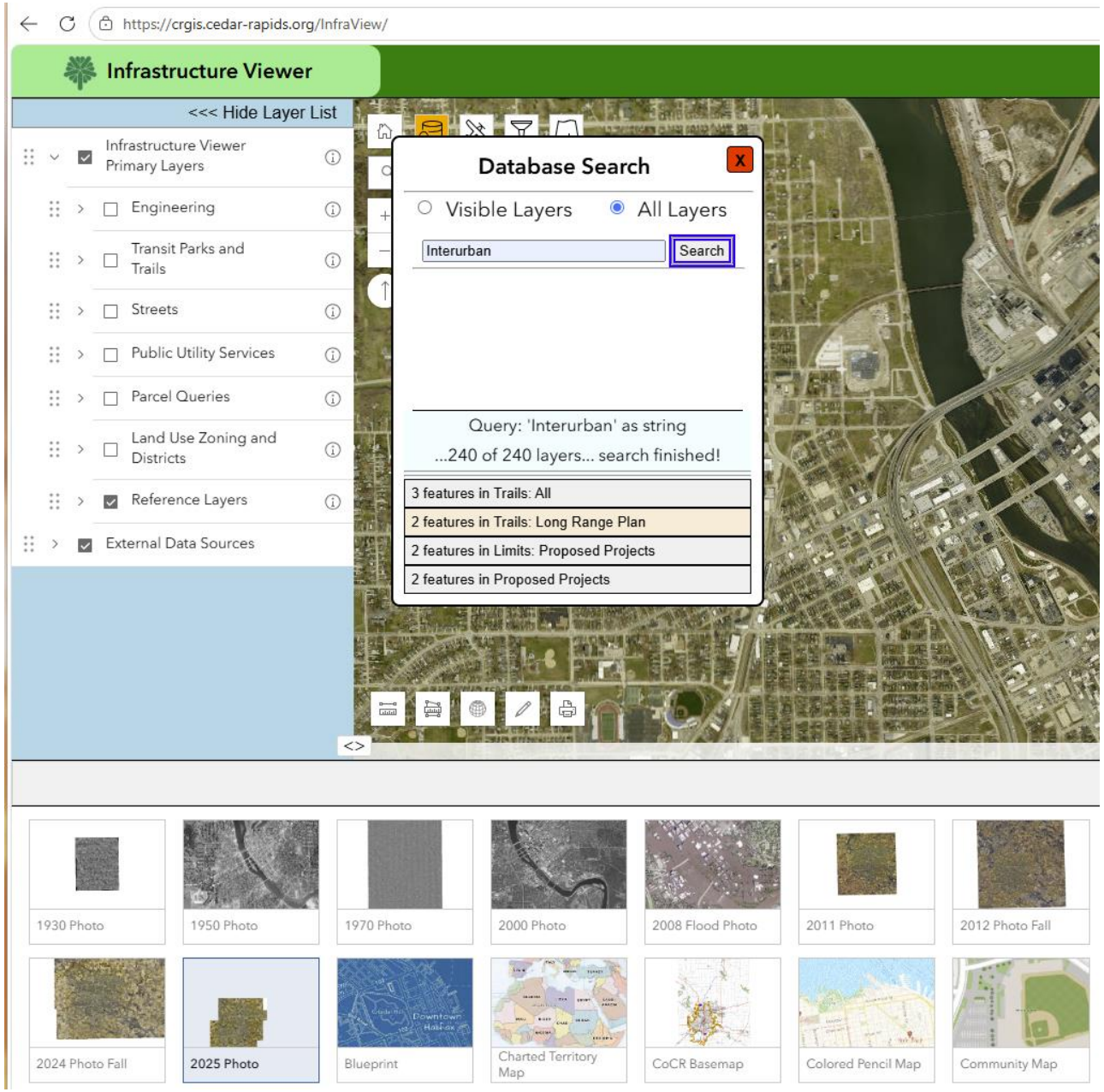
Citation2	*(text)*

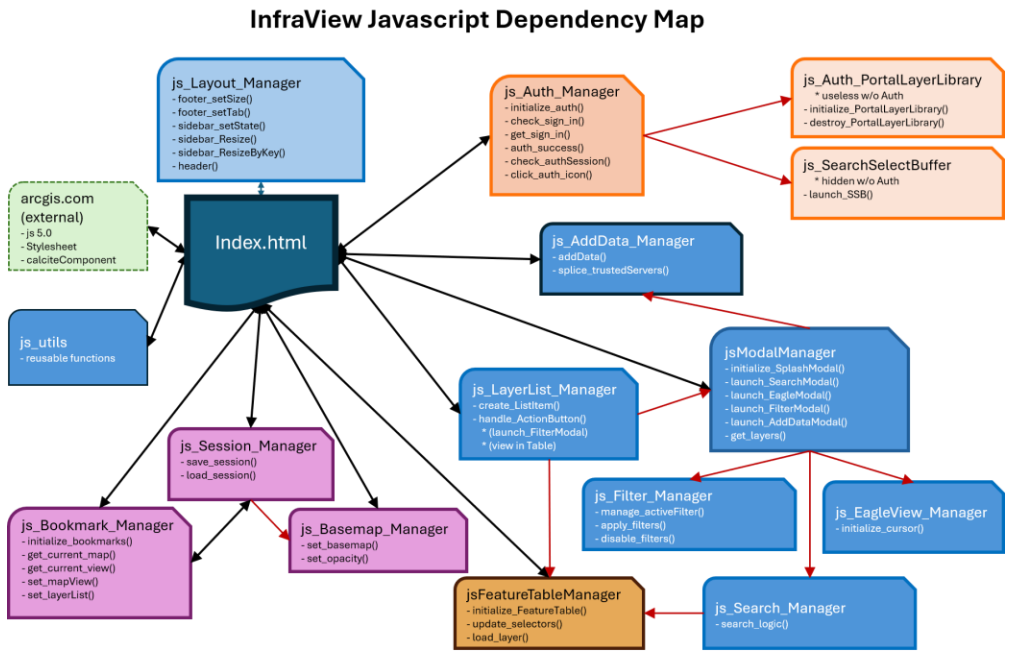
Autofill:

Citation2Date	*(current)*
Citation2By	*(preselected)*
editedDate	*(database)*



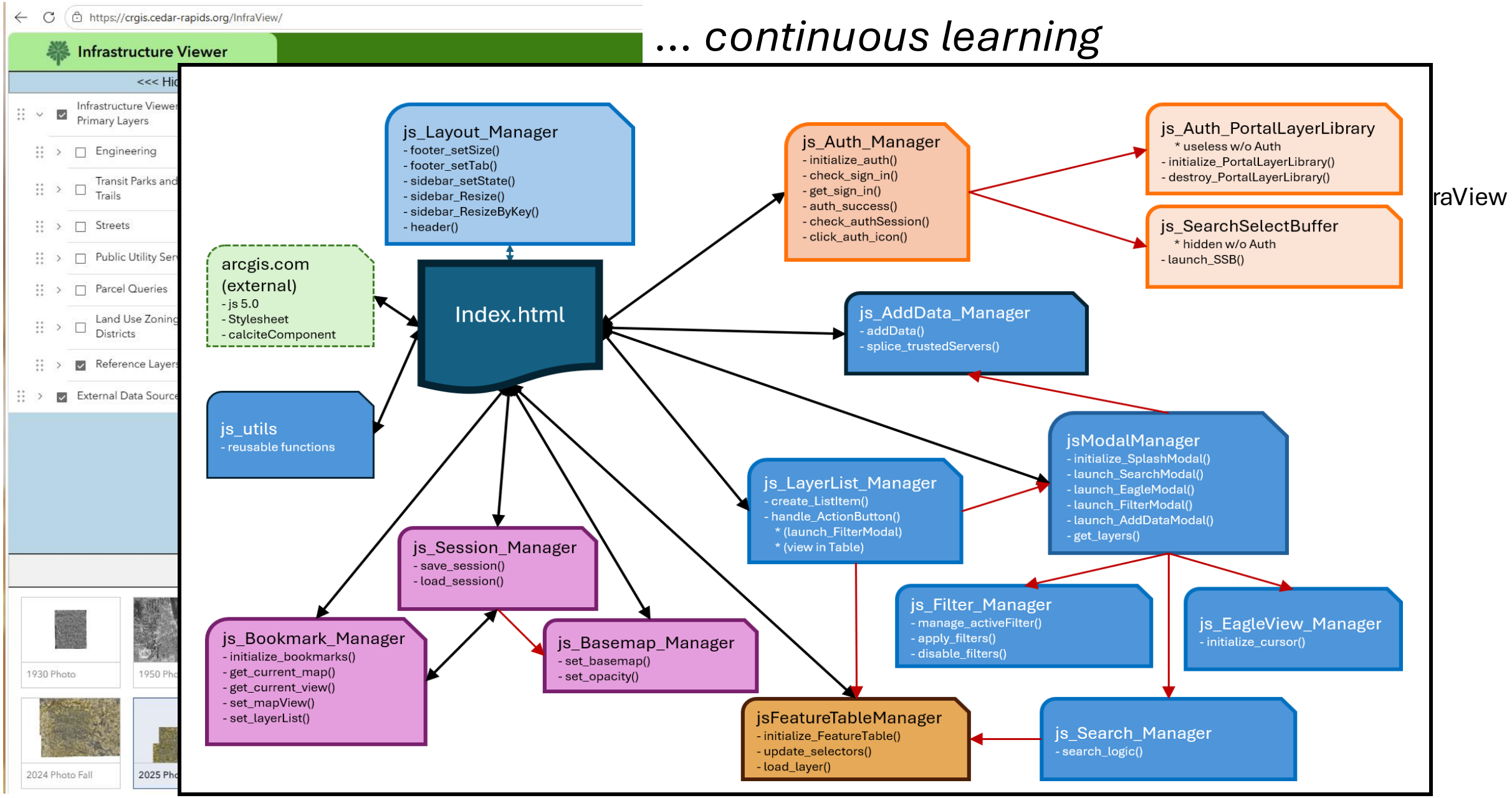
Infrastructure Viewer





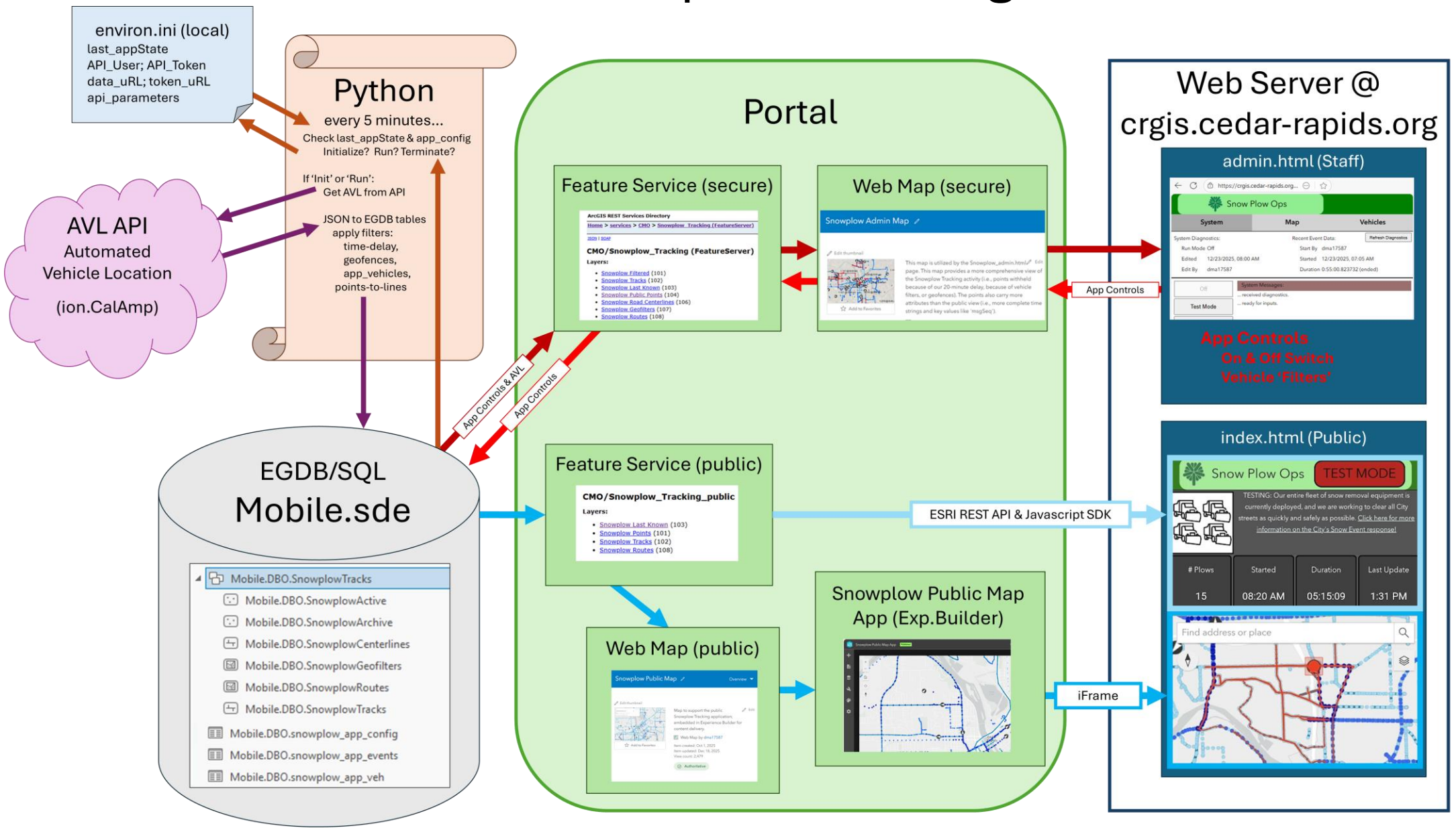
Infrastructure Viewer

... continuous learning



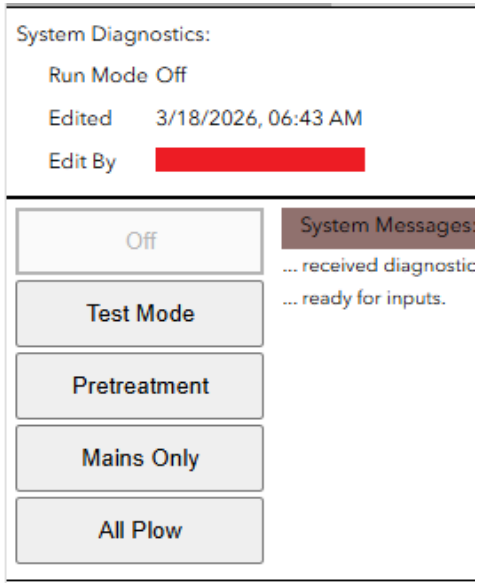
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Snowplow Tracking



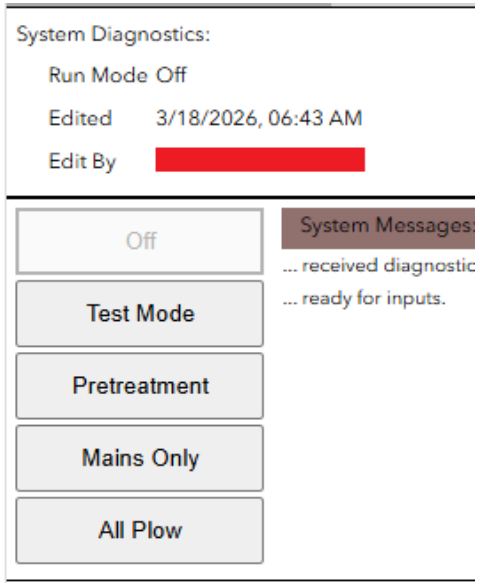
Snowplow Tracking

Workflow Analysis (Ops)

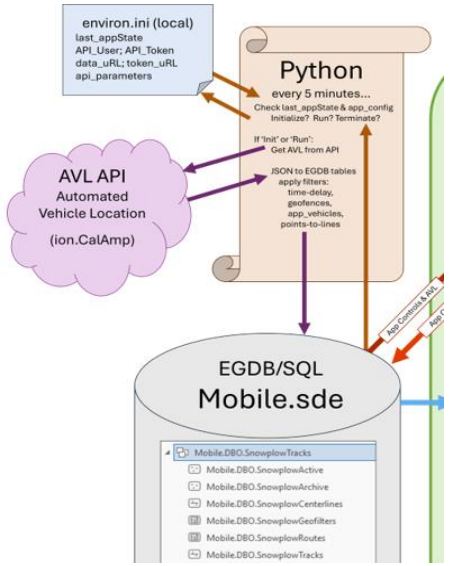


Snowplow Tracking

Workflow Analysis (Ops)

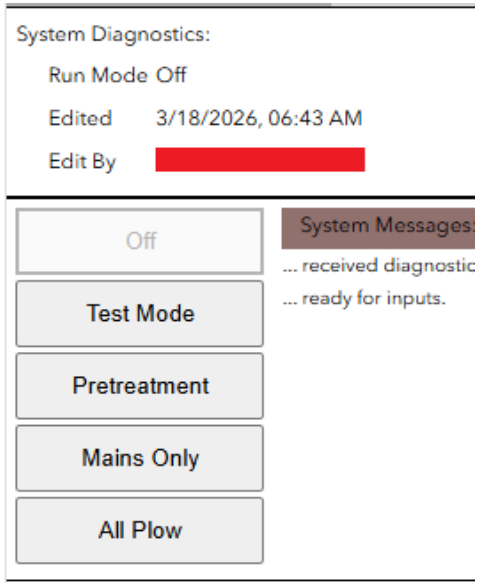


...Continued Learning

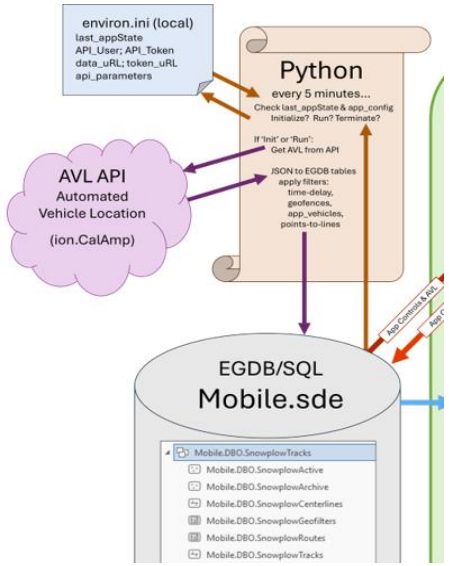


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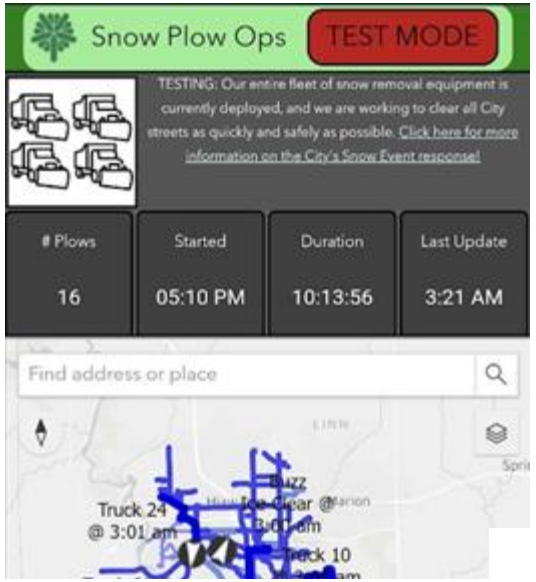
Workflow Analysis (Ops)



...Continued Learning

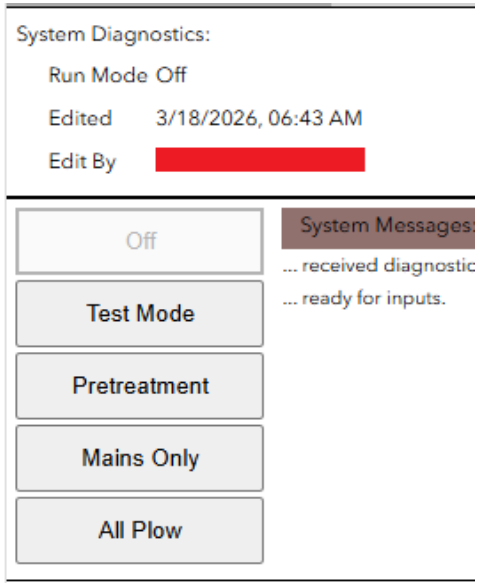


... more Collaboration (Ops & Comms)...

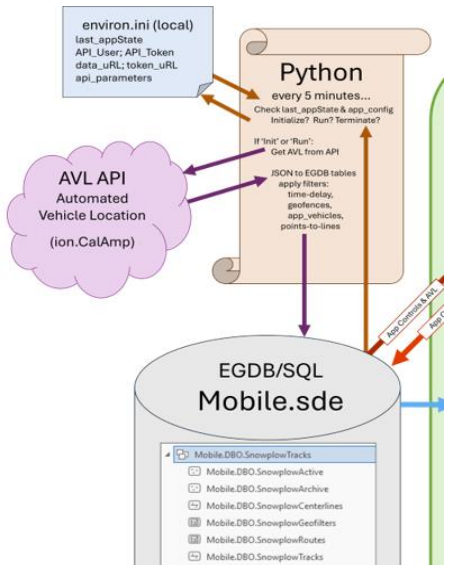


Snowplow Tracking

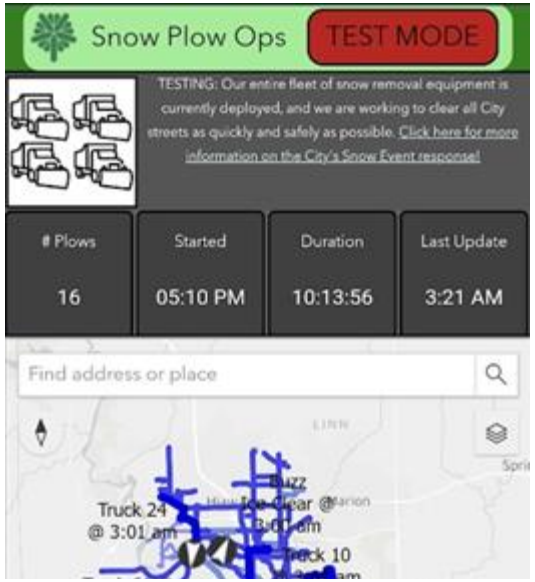
Workflow Analysis (Ops)



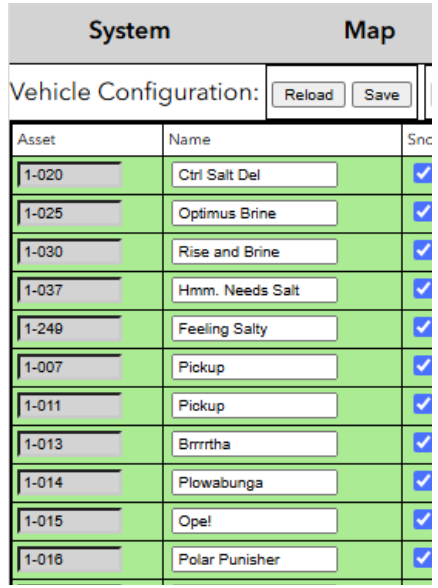
...Continued Learning



... more Collaboration (Ops & Comms)...

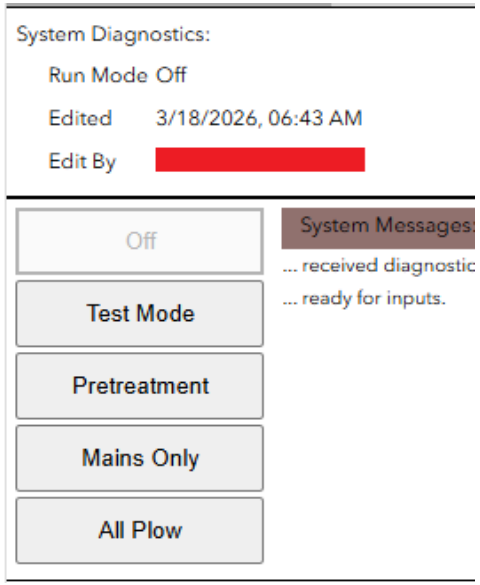


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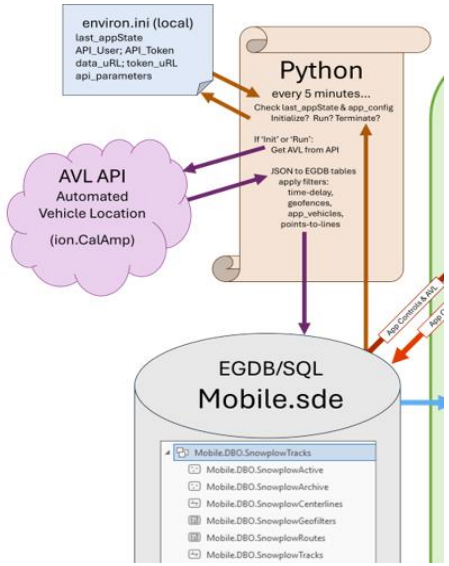


Snowplow Tracking

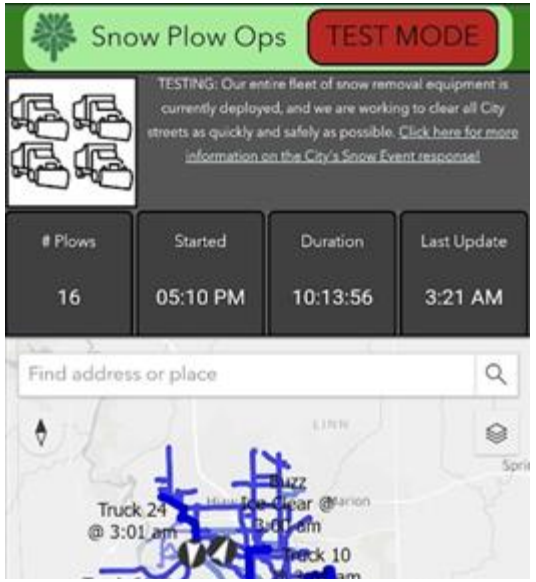
Workflow Analysis (Ops)



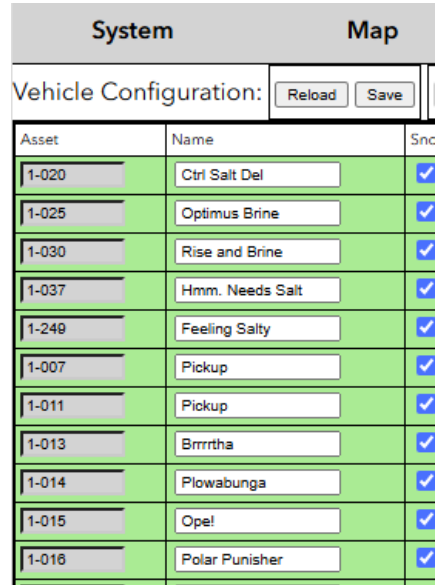
...Continued Learning



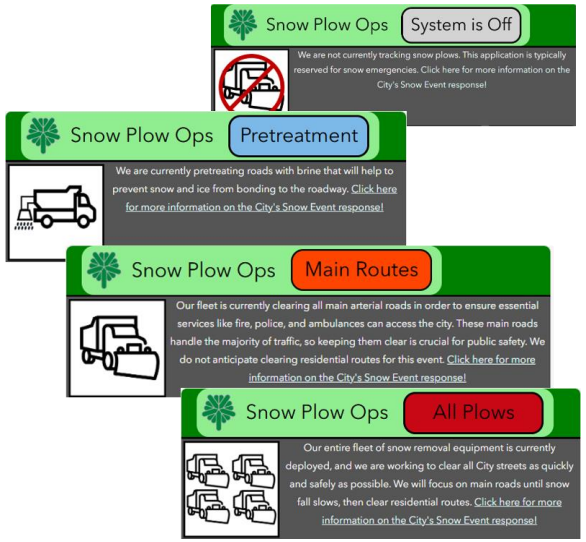
... more Collaboration (Ops & Comms)...



... more collaboration (Comms)

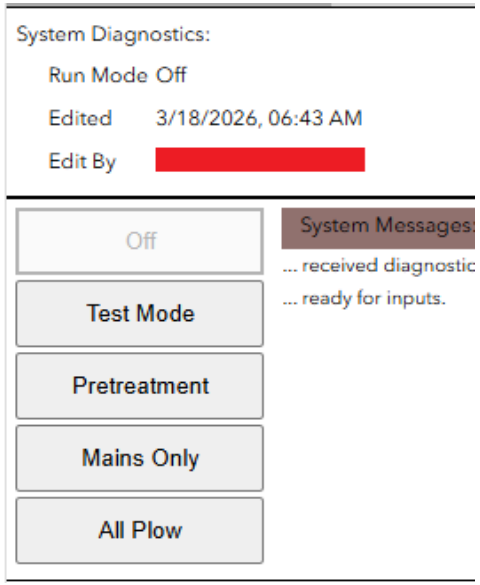


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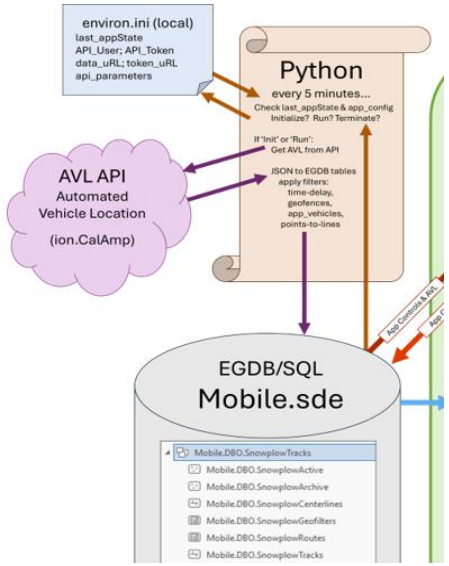


Snowplow Tracking

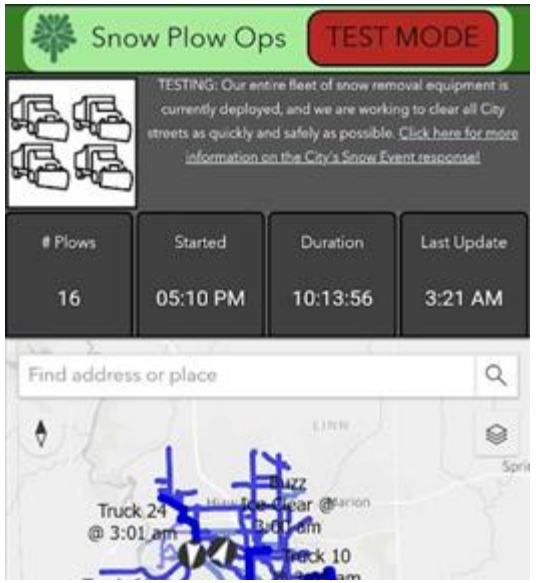
Workflow Analysis (Ops)



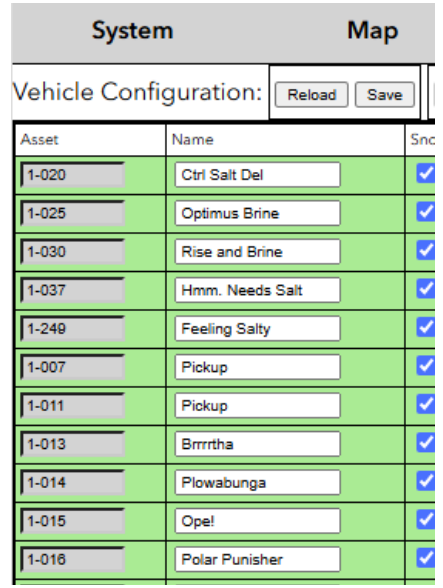
...Continued Learning



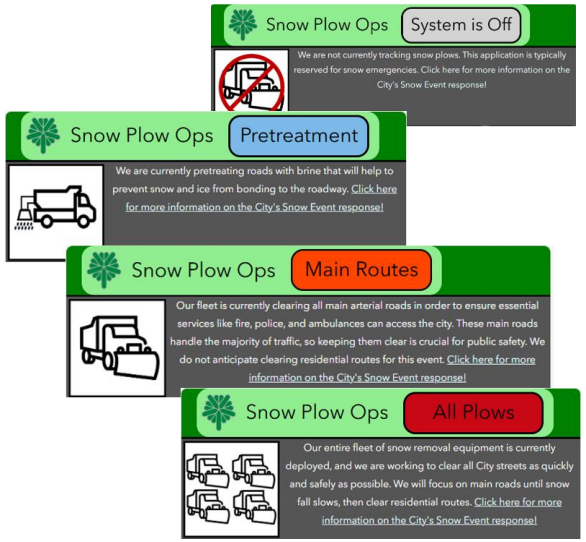
... more Collaboration (Ops & Comms)...



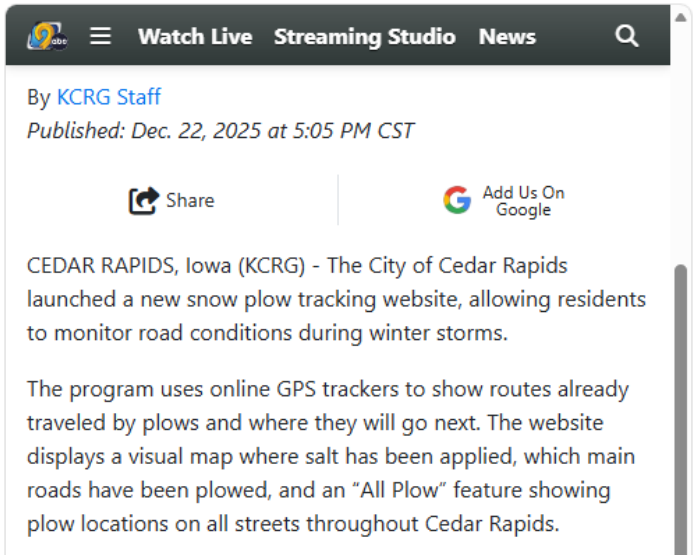
... more collaboration (Comms)



... more collaboration (Ops & Comms)

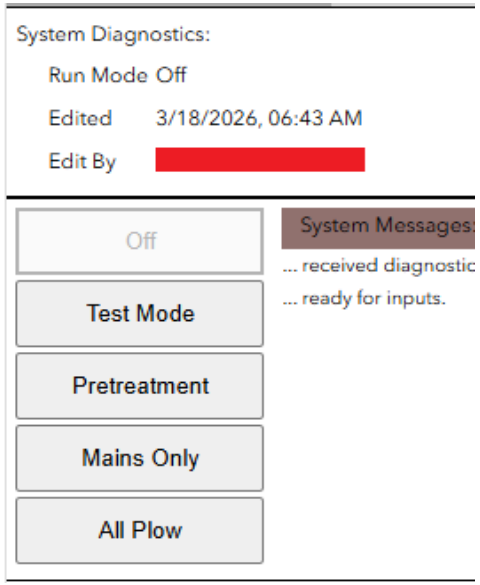


... more collaboration (Comms)

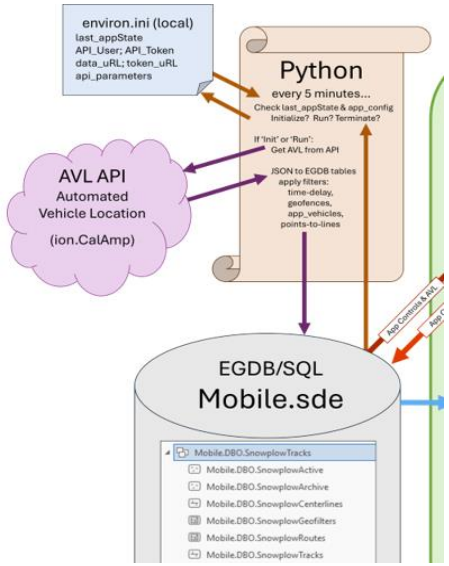


Snowplow Tracking

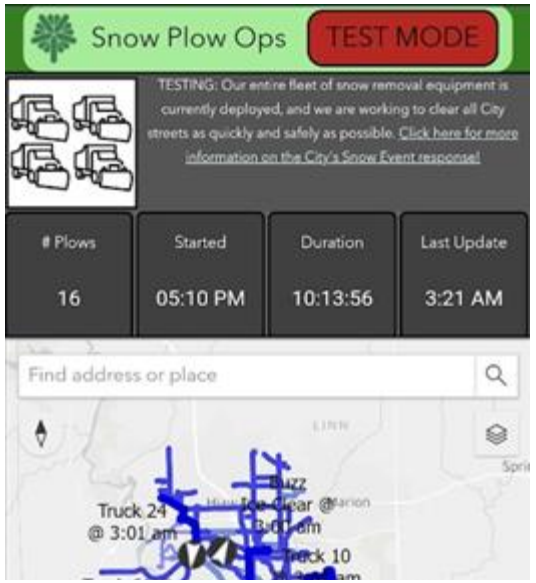
Workflow Analysis (Ops)



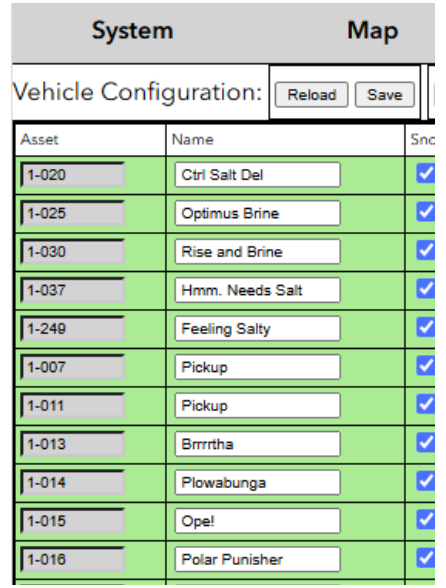
...Continued Learning



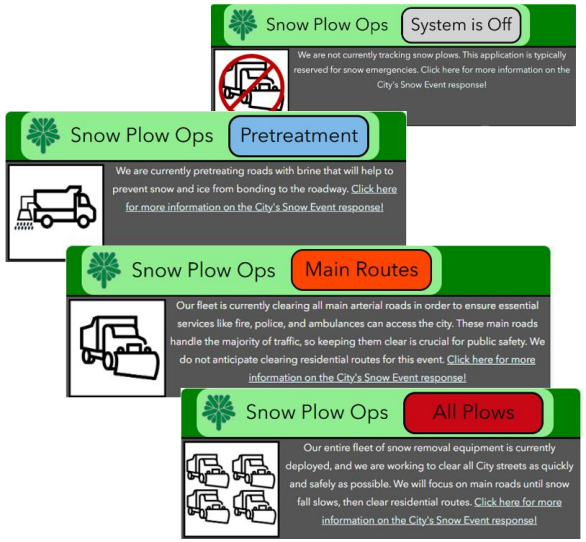
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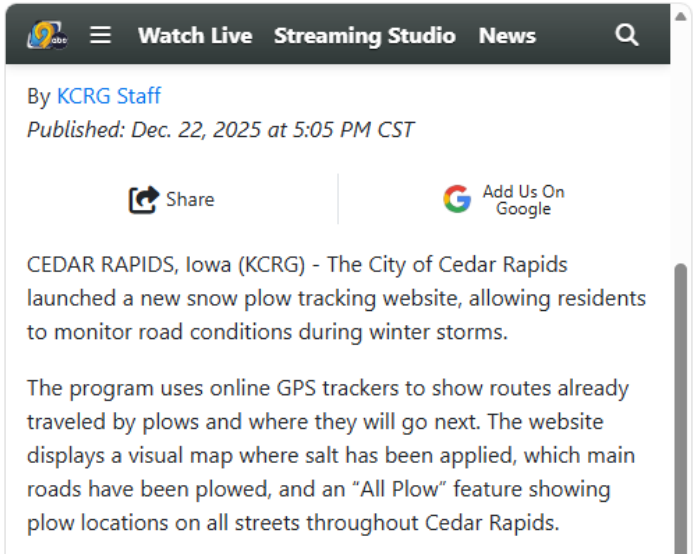
... more collaboration (Comms)



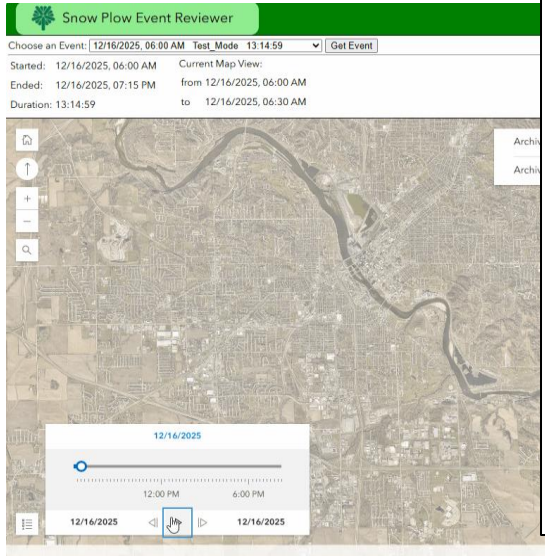
... more collaboration (Ops & Comms)



... more collaboration (Comms)



... more collaboration & workflow (re-)analysis (Ops & CMO)



Event Mileage Report

Event Total Miles: 802.78

Asset	Mileage	GPS Fixes
1-192	121.86	1816
1-016	110.03	1881
1-071	105.13	1835
1-022	96.29	1129
1-185	60.62	948
1-205	57.76	1038
1-091	56.58	963
1-072	53.21	976
1-187	43.99	670
1-097	37.11	703
1-092	25.42	393
1-017	19.24	287
1-041	15.54	216

Conclusions

- Many GIS professionals will encounter an Enterprise GIS in their role.
    - working with an 'eGIS' implies a unique set of challenges *(…which students are not always prepared for…)*
    - but the 'enterprise' also streamlines data management and user administration.
- Technical Skills are essential for success.
    - Python – 'local' processing:
        - eGIS administration, data management, and analysis
    - Javascript – web tools
        - *recommended: ESRI SDK*
        - *(HTML & CSS also assumed)*

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  - working with an 'eGIS' implies a unique set of challenges *(…which students are not always prepared for…)*
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- Technical Skills are essential for success.
  - Python – 'local' processing:
    - eGIS administration, data management, and analysis
  - Javascript – web tools
    - *recommended: ESRI SDK*
    - *(HTML & CSS also assumed)*
- Practice and emphasize Professional Skills
  - Collaboration:
    - few roles are independent or autonomous
    - Collaboration is the first step towards Leadership.
  - Workflow Analysis:
    - Most organizations already have ongoing processes in place.
    - Improving processes begins with shared understanding…
  - Continuous Learning:
    - For GIS, and IT generally… there is no other choice.

Thank you!

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